

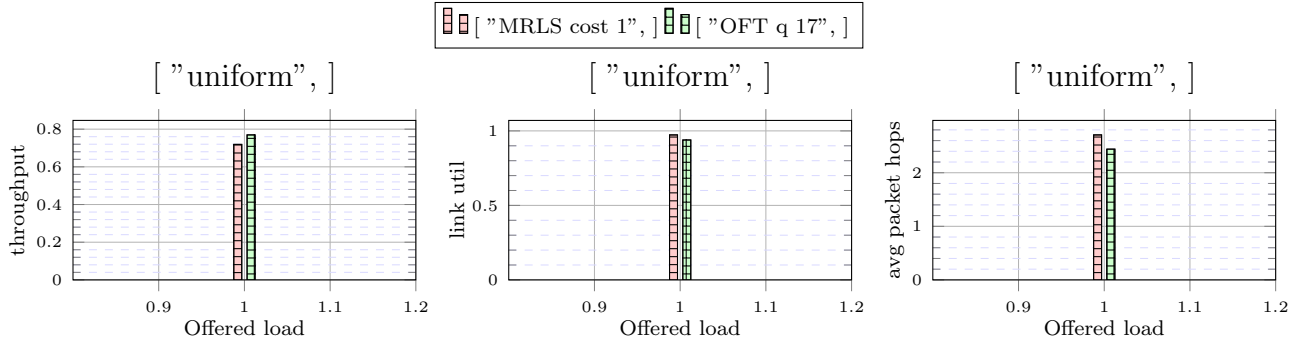


Figure 1: X: ["uniform",]

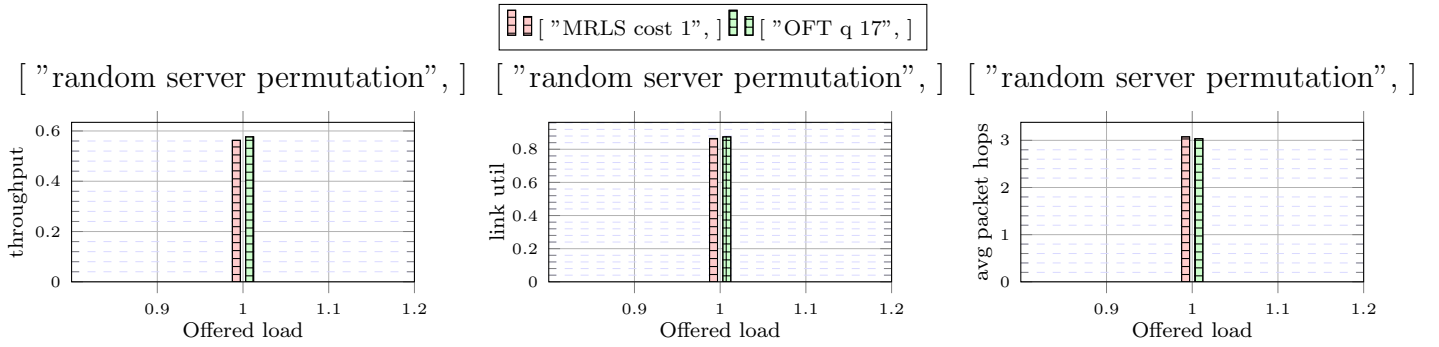


Figure 2: X: ["random server permutation",]

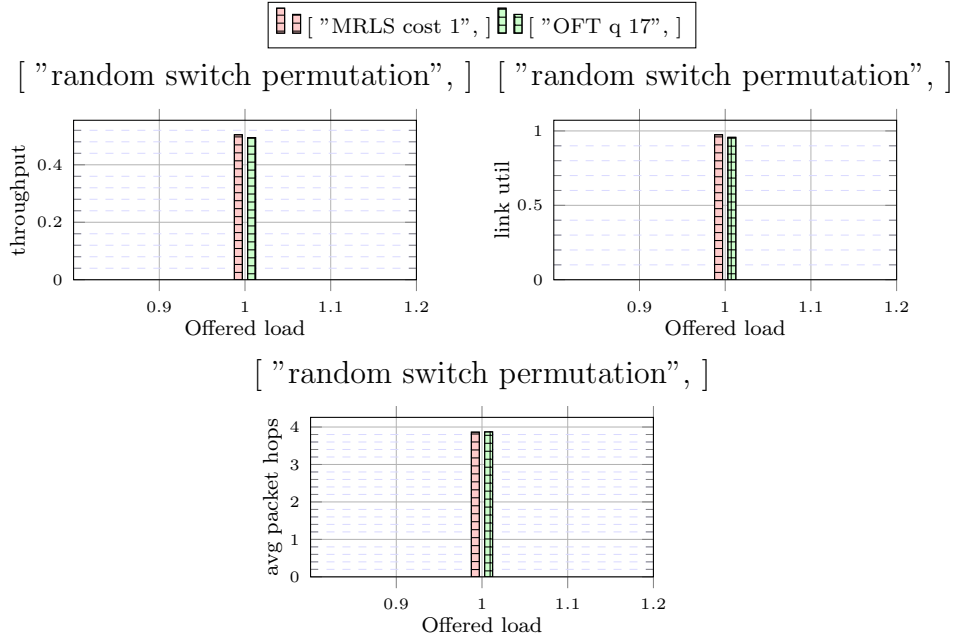


Figure 3: X: ["random switch permutation",]

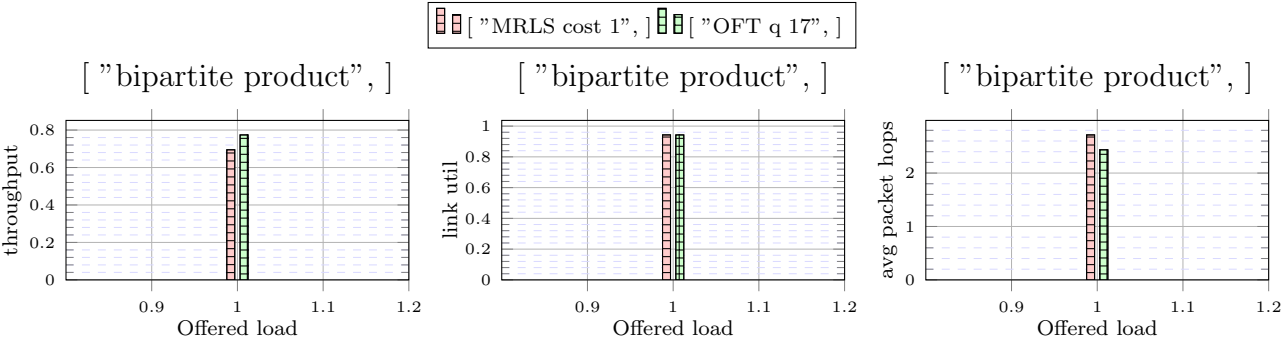


Figure 4: X: ["bipartite product",]

The following versions used in the simulations.

- heads/master-dirty-ce88a576295ce68f40736599a9e3e04f24f38801(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  general_frequency_divisor: 2,
  topology: ![
    MultiStage{
      stages: [
        ExplicitStageFile{ filename: "../MRLS_614_18_307_36" },
        servers_per_leaf: 18,
        legend_name: "MRLS cost 1",
        OFT{ height: 1, prime: 17, servers_per_leaf: 18, legend_name: "OFT q 17" },
        traffic: HomogeneousTraffic{
          pattern: ![
            Uniform{ legend_name: "uniform" },
            RandomPermutation{ legend_name: "random server permutation" },
            CartesianTransform{
              sides: [ 18, 614 ],
              patterns: [
                Identity,
                RandomPermutation{ seed: 500 },
                legend_name: "random switch permutation" },
            Product{
              block_size: 5526,
              block_pattern: Uniform,
              global_pattern: CartesianTransform{
                sides: [ 2 ],
                shift: [ 1 ],
                legend_name: "" },
              legend_name: "bipartite product" },
            servers: 11052,
            load: ![ 0.5 ],
            message_size: 16,
            maximum_packet_size: 16,
            router: InputOutput{
              virtual_channels: 4,
              virtual_channel_policies: [
                MapHop{
                  hop_to_policy: [
                    ArgumentVC{
                      allowed: [ 0 ] },
                    ArgumentVC{
                      allowed: [ 0 ] },
                    ArgumentVC{
                      allowed: [ 1 ] },
                    ArgumentVC{
                      allowed: [ 1 ] },
                    ArgumentVC{
                      allowed: [ 2 ] },
                    ArgumentVC{
                      allowed: [ 2 ] },
                    ArgumentVC{
                      allowed: [ 3 ] },
                    ArgumentVC{
                      allowed: [ 3 ] } ],
                VecLabel{
                  label_vector: [ 0, 64, 80 ] },
                OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
                use_neighbour_space: true, aggregate: true },
                OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
                use_neighbour_space: true, aggregate: false },
                LowestLabel,
                EnforceFlowControl,
                Random ],
              allocator: Random,
              buffer_size: 128,
              bubble: false,
              flit_size: 16,
              intransit_priority: false,
              allow_request_busy_port: true,
              output_buffer_size: 64,
              crossbar_frequency_divisor: 1,
              crossbar_delay: 2 },
            routing: ![
              Polarized{ include_labels: true, strong: false, legend_name: "Polarized" },
            link_classes: [
              LinkClass{ delay: 2 },
              LinkClass{ delay: 2 } ],
            launch_configurations: [
              Slurm{ job_pack_size: 1, time: "3-00:00:00" } ] }
```